

Student Number: _____



ROSEVILLE COLLEGE

MATHEMATICS

2003

PRELIMINARY EXAMINATION

General Instructions

Time Allowed – 3 hours, plus 5 minutes reading time

Directions to Candidates:

- Attempt all questions.
- Begin each question on a new page, with your number clearly written on every page.
- There are 10 questions, each worth 12 marks.
- Individual marks for each question are indicated on the left hand side of each question
- All necessary working must be shown.
- Marks may not be awarded for careless or badly arranged work.
- Approved calculators may be used.

Question 1

Basic Arithmetic (*Start work on a new page*)

(a) Simplify $|-5| - |8|$ 1

(b) Calculate correct to two decimal places 2

$$\frac{34.2}{16.3 \times 2.7}$$

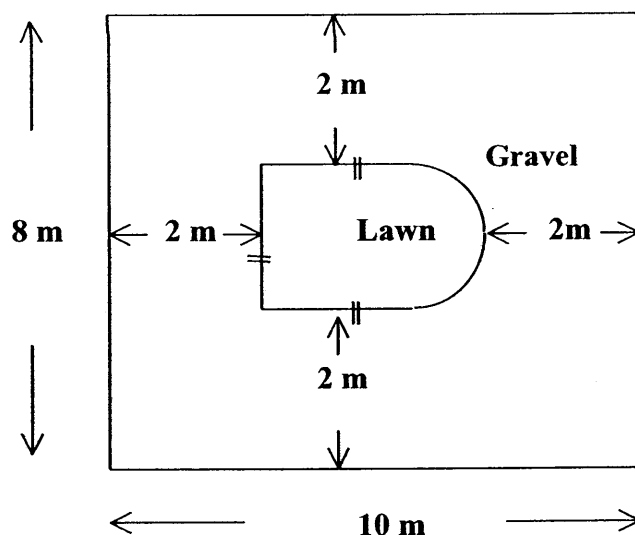
(c) A skirt is priced at \$82.50, including GST of 10%. What was the price of the skirt prior to adding GST? 1

(d) Evaluate $(5.9 \times 10^{55})^2$ in scientific notation, correct to 2 significant figures 2

(e) Find the exact value of $81^{-\frac{1}{4}} \div 2^{-2}$ 1

(f) Express $0.\dot{1}0\dot{2}$ as a fraction in it's simplest form. 2

(g) The Backyard Force TV team is renovating a backyard to the plan shown below. Calculate the area of lawn in the backyard (correct to 1 decimal place) 3

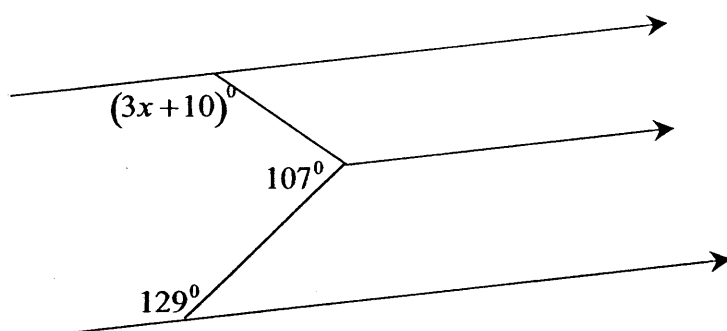


Question 2	Algebra and Surds (Start work on a new page)	Marks
(a)	Simplify $4 - 6(m - 2)$	1
(b)	If $S = \frac{a}{1-r}$, find S when $a = 0.8$ and $r = \frac{1}{2}$	1
(c)	Factorise completely $4a^2 - b^2 + 2a - b$	2
(d)	Simplify $\sqrt{75} + \sqrt{48}$	2
(e)	Express $\frac{2\sqrt{3}-1}{5-\sqrt{3}}$ with a rational denominator.	2
(f)	Simplify $\frac{2}{x(x-3)} - \frac{1}{x}$	2
(g)	Find the exact value of $\frac{t^3}{5} - t^2 + 1$ when $t = -3$	2

Question 3	Equations (Start work on a new page)	Marks
(a)	Solve:	1
i.	$3(2x - 3) = 4(x - 2)$	2
ii.	$\frac{5x}{2} - \frac{2x}{3} = 2$	2
iii.	$x^2 - 3 = 3x + 1$	2
(b)	Mark on the number line the values of x for which $ x - 1 \leq 2$	2
(c)	Find the values of a and b such that :	2
	$\begin{cases} 2a - 3b = 7 \\ a - 4b = 1 \end{cases}$ and	3
(d)	Solve for x : $(x + 1)^2 = 6$, leaving your answer in exact form.	

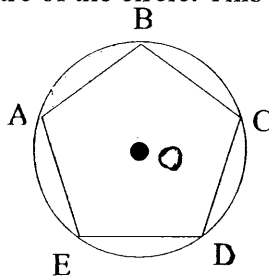
Question 4 **Geometry** (Start work on a new page)

- (a) Find the value of x in the following diagram.

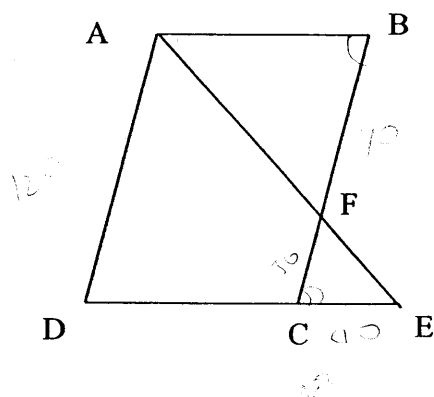


Marks

- (b) ABCDE is a regular pentagon where ABCDE lie on the circumference of a circle of diameter 32 cm. O is the centre of the circle. This is shown below.



- | | |
|--|---|
| (i) Find the length of OA. | 1 |
| (ii) Find the angle AOB. | 1 |
| (iii) Find the angle OAB. | 1 |
| (iv) Find the length of the side of the pentagon, giving your answer to one decimal place. | 2 |
| (v) Find the area of the pentagon, giving your answer to one decimal place. | 1 |
| (vi) Find the area of the sector AOB, giving your answer to one decimal place. | 1 |
- (c) ABCD is a parallelogram with DC produced to E, where $AD = 120\text{mm}$, $CE = 40\text{mm}$ and $BF = 70\text{mm}$.
- (i) Show that $\triangle ABF$ is similar to $\triangle ECF$
- (ii) Find the length of AB



Question 5

Functions and Graphs (Start on a new page)

Marks

(a) A function $f(x)$ is defined as: $f(x) = \begin{cases} 2-x & \text{for } x < -3 \\ 5 & \text{for } -3 \leq x < 0 \\ x^2 - 1 & \text{for } x \geq 0 \end{cases}$

(i) Calculate the value of (a) $f(-10)$

1

(β) $f(-3)$

1

(γ) $f(a^2)$

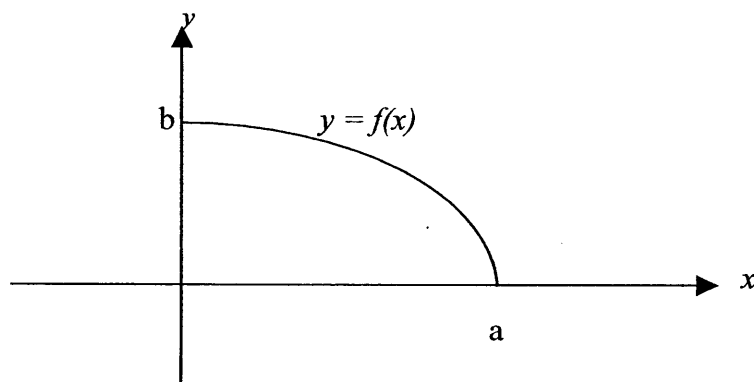
1

(ii) Sketch the graph of $y = f(x)$ over the domain $-5 \leq x \leq 2$, showing all essential features including any intercepts with the x and y axes.

2

(b) The diagram below shows part of the graph of the function $y = f(x)$. You are told that it is an odd function. Copy the diagram and complete the graph of the function.

1



(c) Solve simultaneously:

2

$$\begin{cases} y = 4x + 3 \\ y = x^3 + 3 \end{cases}$$

(d) (i) Sketch the curve $y = x^2 - 2x$, clearly showing all intercepts.

1

(ii) What is the minimum value of this function?

1

(e) Sketch the graph of the function $y = \frac{1}{x+1}$ and state the domain and the range of the function.

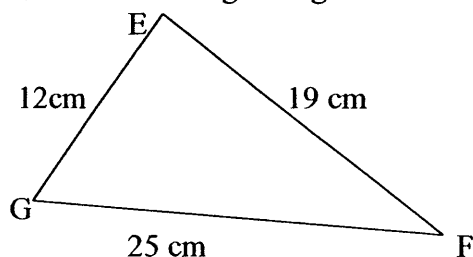
2

Question 6

Trigonometry (Start work on a new page)

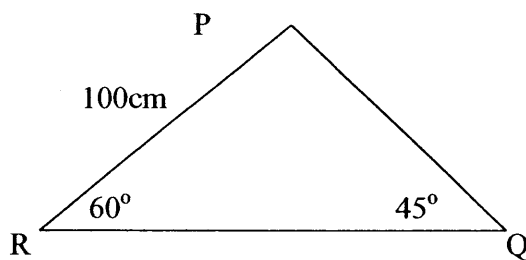
- (a) Find the size of the largest angle in the triangle below, correct to the nearest degree.

2



NOT TO SCALE

- (b) In $\triangle PQR$ below:



- (i) Show that the length of PQ is $50\sqrt{6}$ cm.
- (ii) Hence or otherwise, find the area of the triangle PQR. (correct to 1 decimal place)
- (c) Find all values of θ such that $\cos\theta = \frac{-1}{2}$ and $0^\circ \leq \theta \leq 360^\circ$

2

2

2

- (d) (i) Show that: $\tan \theta + \cot \theta = \frac{1}{\sin \theta \cos \theta}$

1

- (ii) Hence or otherwise, solve:

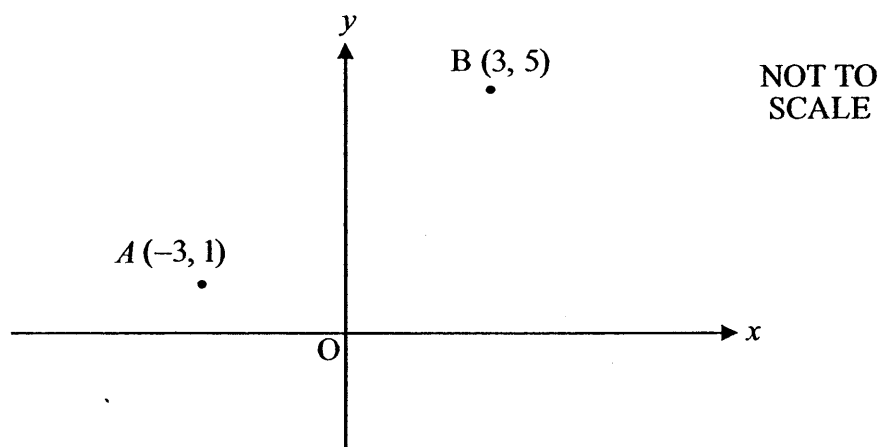
3

$$\frac{1 + \cot \theta}{\operatorname{cosec} \theta} - \frac{\sec \theta}{\tan \theta + \cot \theta} = 1, \quad 0^\circ \leq \theta \leq 360^\circ$$

Question 7 **Straight Line Graphs** (*Start work on a new page*)

- (a) Sketch the graph of $y = -x + 2$ on a set of axes, showing any x and y intercepts.

2



- (b) The diagram shows the points $A(-3, 1)$ and $B(3, 5)$ on the Cartesian number plane.

Copy or trace this diagram onto your writing page.

- (i) Show that the equation of AB is $2x - 3y + 9 = 0$.
- (ii) Show that the point C , which is the midpoint of AB is the y -intercept of AB .
- (iii) Calculate the perpendicular distance from the point $D(2, 0)$ to the line AB and mark the point D on your diagram.
- (iv) The point E , lies on the line $y = -1$ and the line BE is perpendicular to the line AB . Show that E has the coordinates $(7, -1)$ and mark point E on your diagram.
- (v) Show that $BCDE$ is a trapezium.
- (vi) Find the area of $BCDE$.

2

1

2

2

1

2

Question 8	Introductory Calculus (<i>Start work on a new page</i>)	Marks
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(a) Differentiate with respect to x .

(i) $2x^4 + 7x^2 + 7$

1

(ii) $x^{\frac{1}{3}}$

1

(iii) $\sqrt{x^2 + 4}$

1

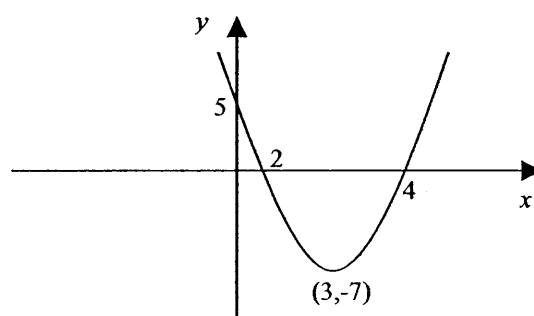
(iv) $\frac{6x^2 - 7}{3 - 2x}$

2

(b) The graph shown below represents $f(x)$

Sketch a graph to show the general shape of $f'(x)$.

1



(c) Find the equation to the normal to the curve $y = x^2 + x$ at the point where $x = 1$.

2

(d) A tangent is drawn to the parabola $y = x^2 - 4x$ at the point P. The tangent has a gradient of 6.

(i) Show that the coordinates of P are (5, 5).

2

(ii) Find the equation of the tangent at P.

1

(iii) Find the gradient of the normal at P.

1

Question 9

Quadratic Functions (Start work on a new page)

Marks

- | | | |
|-------|--|---|
| (a) | Find the value of k for which $x^2 - (k-1)x - k = 0$ has equal roots | 2 |
| (b) | Solve for x : $x^6 - 7x^3 - 8 = 0$ | 2 |
| (c) | Find the values of P , Q and R if $3x^2 + 5x - 1 \equiv P(x+1)^2 + Q(x+1) + R$ | 3 |
| (d) | For what values of x is $x^2 + 5x > 0$? | 2 |
| (e) | For the quadratic expression: $2x^2 - x + 3$ which has zeros when $x = \alpha$ and $x = \beta$. | |
| (i) | Show that the expression is positive definite. | 1 |
| (ii) | Find the value of $\alpha\beta$. | 1 |
| (iii) | Find the value of $\frac{1}{\alpha} + \frac{1}{\beta}$. | 1 |

Question 10	Locus and the Parabola (Start work on a new page)	Marks
(a)	For the parabola $16y = x^2$ write down the coordinates of the focus and the equation of the directrix.	2
(b)	The points P and Q have the coordinates $(-1, 0)$ and $(3, 3)$ respectively. If PR is perpendicular to QR, show that the locus of R is $x^2 + y^2 - 2x - 3y - 3 = 0$	2
(c)	Find the equation of the parabola which has its vertex at $(2, 0)$ and its directrix is given by $x = 5$.	2
(d)	By expressing $y = \frac{x^2}{-16} + \frac{6x}{-16} - \frac{7}{-16}$ in the form $(x - h)^2 = 4a(y - k)$, find:	6
	(i) The focal length	
	(ii) The coordinates of the vertex	
	(iii) The equation of the axis	
	(iv) The equation of the directrix	
	(v) Sketch the graph	

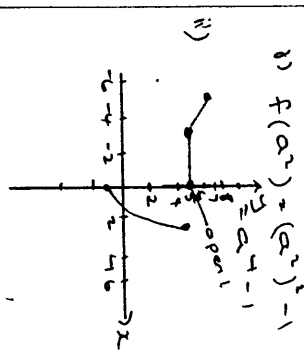
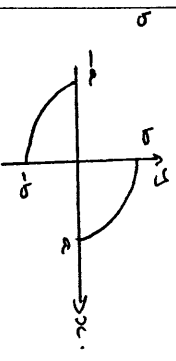
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Suggested Answers

Task Mathematics Preliminary 2003

Question 1		
a. -3	1	2 continued
b. 0.78	2	e. $(2\sqrt{3}-1)(5+\sqrt{3})$
c. 3.75	1	25-3
d. 34.81×10^{110} (without comma!)	1	$= \frac{10\sqrt{3}-5-\sqrt{3}+5}{22}$
$= 3.481 \times 10^{111}$	1/2	$= \frac{9\sqrt{3}+1}{22}$
$= 3.5 \times 10^{111}$ (2.55)	1/2	
e. $\frac{4}{3}$		
f. $x=0.102102...$		
$1000x=102.102...$	1 1/2	f. $\frac{2}{x(x-3)} - \frac{1}{x}$
$999x=102$		$= \frac{2-(x-3)}{x(x-3)}$
$x=\frac{102}{999}$		$= \frac{5-x}{x(x-3)}$
$= \frac{34}{333}$	2	
g. Area = $4 \times 4 + \frac{1}{2} \times \pi \times 2^2$	2	g. $\frac{t^2}{5} - t^2 + 1$
$= 16 + 2\pi$		$= \frac{(-3)^3}{5} - 9 + 1$
$\div 22.3m^2$	3	$= -13\frac{2}{3}$
Question 2		
a. $4-6(m-2)$	1	Question 3
$= 4-6m+12$		a) $3(2x-3)=4(x-2)$
$= 16-6m$		$6x-9=4x-8$
b. $s=\frac{0.8}{1-\frac{1}{2}}$	1	$2x=1$
$= 1.6$		$x=\frac{1}{2}$
c. $(2a-b)(2a+b) + (2a-b)$	1	ii) $\frac{4 \times 5x}{2} - \frac{2x \times 6}{3} \times 6$
$= (2a-b)(2a+b+1)$		$15x-4x=12$
d. $\sqrt{15} + \sqrt{48}$	1	$11x=12$
$= 5\sqrt{3} + 4\sqrt{3}$		$x=\frac{12}{11}$
$= 9\sqrt{3}$		iii) $x^2-3=3x+1$
e) $\frac{2\sqrt{3}-1}{5-\sqrt{3}} \times \frac{5+\sqrt{3}}{5+\sqrt{3}}$	1	$x^2-3x-4=0$
		$(x-4)(x+1)=0$
		$x=4, -1$

Task Question 2 continued

b. $x-1=2$	-1	-1	-1
$x=3$ (i)			
$x=-1$ (ii)			
$x=1$ (iii)			
$ x-1 \leq 2$			
test $x=0$			
$-1 \leq 2$ ✓ true			
$-1 \leq 2$ ✓ true			
c. $2a-3b=7$	2	2	2
$2a-8b=2$			
$5b=5$			
$b=1$			
d. $x+1=\pm\sqrt{6}$			
$x=-1\pm\sqrt{6}$			
$x^2+2x+1=6$			
$x^2+2x-5=0$			
$x=\frac{-2\pm\sqrt{4+4 \times 5}}{2}$			
$=\frac{-2\pm\sqrt{24}}{2}$			
$=\frac{-2\pm 2\sqrt{6}}{2}$			
$=-1\pm\sqrt{6}$			
Question 4			
a) $3x+10+107+129=360^\circ$			
$3x=114$			
$x=38^\circ$			
b) $OA=16cm$			
ii) $\frac{360}{5}=72^\circ$			
$\angle AOB=72^\circ$			
iii) $\angle AOB=54^\circ$			
iv) $x^2=16^2+16^2-2 \times 16^2 \cos 72^\circ$			
$x=18.8cm$			
v) Area = $\frac{1}{2} \times 5 \times 16 \times 16 \sin 72^\circ$			
$=121.7cm^2$			
$\therefore$ Pentagon = $608.7cm^2$			
vi) $A=\frac{\pi}{3} \times 16^2$			
$=168.8cm^2$			
c) $\angle ABE = \angle FCE$ (alt $\angle$ & $\parallel$ lines)			
$\angle AEB = \angle CFE$ (vert. opp)			
$\therefore \triangle ABE \parallel \triangle CFE$ (equiangular)			
ii) $\frac{AB}{40} = \frac{70}{50}$			
$\therefore AB=56mm$			
Question 5			
i. a) $f(-10)=2-(-10)$			
$=12$			
b) $f(-3)=5$			
c) $f(a^2)=(a^2)^2-1$			
$=a^4-1$			
ii) 			
iii) 			

Task Maths Prelim. Suggested answers 2008

Question 6

c. $y = 4x + 3$
 $y = 8x^3 + 3$

$x^3 + 3 = 4x + 3$

$x^3 - 4x = 0$

$x(x+2)(x-2) = 0$

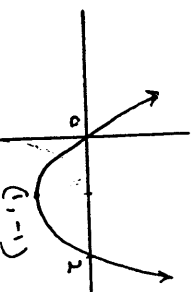
$x = 0, \pm 2$

$x = -2, y = -5$

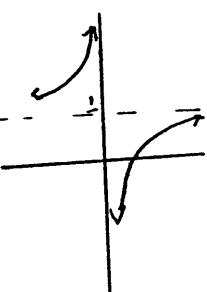
$x = 0, y = 3$

$x = 2, y = 11.$

d. $y = x^2 - 2x$
 $= x(x-2)$



min value is -1



Domain all $x, y \neq -1$
 Range all y , except $y \neq 0$

Question 7

a. $\cos \theta = \frac{12^2 + 19^2 - 25^2}{2 \times 12 \times 19}$

$\theta = 105^\circ$

i) $\frac{\sin \theta}{\cos \theta} + \frac{\cos \theta}{\sin \theta}$

$= \frac{\sin^2 \theta + \cos^2 \theta}{\sin \theta \cos \theta}$

$= \frac{1}{\sin \theta \cos \theta}$

$= \frac{1}{\frac{1}{2} \sin 2\theta}$

ii) $(1 + \frac{\cos \theta}{\sin \theta}) \div \frac{1}{\sin \theta} = \frac{1}{\cos \theta} \times \frac{\sin \theta}{1} = \frac{\sin \theta}{\cos \theta} = \tan \theta$

$= \frac{\sin \theta}{\cos \theta} \times \frac{1}{1} = \frac{\sin \theta}{\cos \theta} = \tan \theta$

$\therefore \cos \theta = 1$

$\theta = 0, 360^\circ$

Sorry about order!

$\frac{x}{\sin 60} = \frac{100}{\sin 45}$

$\frac{x}{\sqrt{3}/2} = \frac{100}{1/\sqrt{2}}$

$2x = 100\sqrt{6}$

$x = 50\sqrt{6}$

ii) $A = \frac{1}{2} \times 100 \times 50\sqrt{6} \sin 75^\circ$

$\approx 5915.1 \text{ cm}^2$

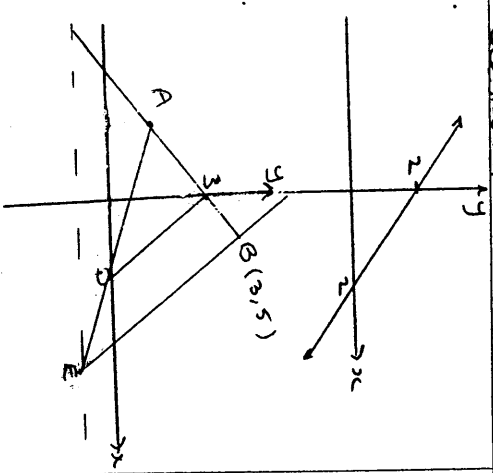
c) $\theta = 60^\circ$ in the 2nd quadrant

$\therefore \theta = 120^\circ, 240^\circ$

Question 7

Task

a.



i) $m = \frac{2}{3}$

$y - 1 = \frac{2}{3}(x + 3)$

$3y - 3 = 2x + 6$

$2x - 3y + 9 = 0$

ii) $m(x, y) = (\frac{2}{3}, \frac{1}{3}) = (0, 3)$

$2x - 3y + 9 = 0$

at $x = 0$

$0 - 3y + 9 = 0$

$\therefore y = 3.$

$\therefore$ the midpoint of AB is also the y-int. of the line AB

iii) $AD = \frac{|2 \times 1 - 3 \times 0 + 9|}{\sqrt{4 + 9}}$

$= \frac{11}{\sqrt{13}} \times \frac{\sqrt{13}}{\sqrt{13}}$

iv) $m_2 = -\frac{3}{2} = \sqrt{13} \text{ units}$

$y - 5 = -\frac{3}{2}(x - 3)$

$2y - 10 = -3x + 9$

$3x + 2y - 19 = 0$

at $y = -1, 3x - 2 - 19 = 0$

v) $m \text{ of } CD = -\frac{3}{2}$

$m \text{ of } BE = -\frac{3}{2}$

$\therefore CD \parallel BE$

$\therefore BCDE$ is a quadrilateral pair of parallel lines $\therefore$ a trapezium

vi) $BE = \sqrt{16 + 36} = \sqrt{52} = \sqrt{4 \times 13} = 2\sqrt{13}$

$BC = \sqrt{13}$

$\therefore A = \frac{1}{2} \times \sqrt{13} (\sqrt{13} + 2\sqrt{13})$

$= \frac{\sqrt{13}}{2} \times 3\sqrt{13}$

$= \frac{39}{2} \text{ u}^2$

Question 8

i) $8x^3 + 14x$

ii) $\frac{1}{3} x$

iii) $\frac{1}{2} (x^2 + 4) \times 2x$

$= \frac{x}{\sqrt{x^2 + 4}}$

iv) $f(x) = \frac{3x^2 - 7}{3 - 2x}$

$f'(x) = \frac{(3 - 2x)(12x - 2(6x^2 - 7))}{(3 - 2x)^2}$

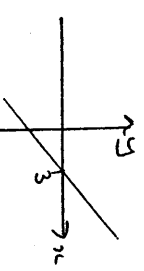
$= \frac{36x - 24x^2 + 12x^2 - 14}{(3 - 2x)^2}$

$= \frac{-12x^2 + 36x - 14}{(3 - 2x)^2}$

$= \frac{(3 - 2x)^2}{(3 - 2x)^2}$

$= \frac{-12x^2 + 36x - 14}{(3 - 2x)^2}$

b.



Question 8 cont.

c. $y = x^2 + 2x$
 $y' = 2x + 1$
 at $x=1, y=2$

$\therefore m_1 = 3$
 $m_2 = -\frac{1}{3}$
 $y - 2 = -\frac{1}{3}(x - 1)$

$3y - 6 = -x + 1$
 $\therefore x + 3y - 7 = 0$
 or $y = -\frac{1}{3}x + \frac{7}{3}$

d(i) $y = x^2 - 4x$
 $y' = 2x - 4$
 at $y' = 0$
 $2x - 4 = 0$
 $2x = 4$
 $x = 2, y = 25 - 20 = 5$

$\therefore P$ is $(5, 5)$
 $y - 5 = 6(x - 5)$
 $y - 5 = 6x - 30$
 $y = 6x - 25$

(ii) $m_2 = -\frac{1}{6}$

Question 9

a) $x^2 - (x-1)x - k = 0$
 $\Delta = 0$
 $(x-1)^2 + 4k = 0$
 $x^2 - 2x + 1 + 4k = 0$
 $x^2 + 2k + 1 = 0$
 $(k+1)^2 = 0$
 $\therefore k = -1$

b. $x^2 - 7x^3 - 8 = 0$
 $(x^3 - 8)(x^2 + 1) = 0$
 $\therefore x^3 = 8, -1$

$x = 2, x = -1$
 $c. 3x^2 + 5x - 1 = P(x+1)^2 + Q(x+1) + R$
 $= Px^2 + 2Px + P + Qx + Q + R$
 $= Px^2 + (2P+Q)x + P+Q+R$

$\therefore P=3, 2P+Q=5, P+Q+R=-1$
 $Q=-1, R=-3$

d. $x^2 + 5x > 0$
 $x(x+5) > 0$
 $x < -5, x > 0$

e) $\Delta = 1 - 11, a = 2$
 $= -10$
 since $\Delta < 0$ & $a > 0$,
 $\Rightarrow x$ is definite

ii) $aP = \frac{a}{a} \quad \text{iii) } \frac{1}{a} + \frac{1}{b}$
 $= \frac{3}{a}$
 $= \frac{a+b}{ab}$
 $= \frac{\frac{1}{3} + \frac{1}{3}}{\frac{1}{3} \cdot \frac{1}{3}} = \frac{\frac{2}{3}}{\frac{1}{9}} = \frac{2}{3} \cdot \frac{9}{1} = 6$

Question 10

a) focus $(0, 4)$
 directrix $y = -4$

b) $PR = \frac{y-0}{x+1} \quad (t) \quad QR = \frac{y-3}{x-3} \quad (t)$

$\frac{y}{x+1} = -\frac{(x-3)}{y-3}$
 $y(y-3) = -(x-3)(x+1)$
 $y^2 - 3y = -(x^2 - 3x + x - 3)$
 $y^2 - 3y = -x^2 + 2x + 3$
 $\therefore x^2 + y^2 - 2x - 3y - 3 = 0$

Question 10 cont.

c) $(y-k)^2 = 4a(x-h)$
 $(y-0)^2 = -4 \times 3(x-2)$
 $y^2 = -12(x-2)$
 $h=2, k=0$

d) $-16y = x^2 + 6x - 7$
 $x^2 + 6x + 9 = -16y + 7 + 9$
 $(x+3)^2 = -16(y-1)$

i) $a = 4$ (not -4)
 ii) $(-3, 1)$
 iii) $x = -3$
 iv) $y = 5$

